Histology Written

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وَمَا تَوْفِيقِى إِلَّا بِاللَّهِ

استعن بالله ولا تعجز

اللَّهمَّ لا سَهلَ إلَّا ما جَعَلتَه سَهلًا، وأنتَ تَجعَلُ الحَزْنَ إذا شِئتَ سَهلًا

ملحوظة : هذه المذكرة شغل خاص بتيم الحريف ولا نسامح فى سرقتها

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Histology and its methods of study

Complete

…… is the increase in linear dimensions of an object without optical defects

- Magnification

……..is the power of the microscope to distinguish fine details

- Resolution

……..is the shortest distance between two details below which they appear as one or cannot be seen at all with the microscope

- limit of resolution

…….microscope in which the day or artificial lamp light is used for ……

- ordinary light –illumination

……microscopy permits direct observation of living cells

- Phase contrast

substance appears brilliant on a dark background in ……. microscope

- Fluorescent

…….florescent stain which can combine with DNA and RNA

- acridine orange

……..electron microscopes used to examine the fine details of the cells and tissues

- Transmission

…… electron microscope used to study the details of surfaces and contours of the cells

- Scanning

limit of resolution in ……microscope is 0.24 nm but in ……. microscope is 0.24 um

- electron—light

in tissue culture , piece of tissue is treated with…… to dissociate its components and the cells

- trypsin

in paraffin section ,, fixation of tissue by…….. , Dehydration by using ascending grades of …… , Clearing by……..

- formalin—alcohol—xylol

in electron microscopic Preparation ,, fixation of tissue by…….. followed by…… , Dehydration by using ascending grades of …… , Clearing by……..

- gluteraldehyde – osmic acid –alcohol—propylene oxide

in paraffin section ,, …… is used for cutting but in electron microscopic Preparation…….. is used

- microtome—ultra-microtome

……is used in staining in electron microscopic Preparation

- uranyl acetate

…….. is a Special tissue preparations used in examination of tissue fluids

- Filming

…….. is a Special tissue preparations used in examination of blood

- Filming

…….. is a Special tissue preparations used in examination of sticky secretions

- Smearing

…….. is a Special tissue preparations used in examination of bone marrow

- Smearing

…….. is a Special tissue preparations used in examination of fibers

- Teasing

…….. is a Special tissue preparations used in examination of muscle fibers or nerve fibers

- Teasing

…….. is a Special tissue preparations used in examination of thin tissue or connective tissue

- Spreading

…….. is a Special tissue preparations used in examination of bone

- Grinding

……stain is a basic dye that has the ability to stain acidic structures in the cell

- Hematoxylin

……is an acidic dye that has the ability to stain basic structures in the cell

- Eosin

The general stain is…….and …….

- Hematoxylin and Eosin

Enumerate

The two main functions of microscope or What are the functions of microscope {سؤال في المحاضرة}

- Magnification
- Resolution

Types of microscopes {أسئلة كتاب القسم}

- **ordinary light** microscope

- **Phase contrast** microscopy
- **Fluorescent** microscopy
- **Electron** microscopy

Types of electron microscopes

- Transmission electron microscopes
- High voltage electron microscope
- Scanning electron microscope

Two uses of Examination of living cells (tissue culture)

- Study the effect of a toxic substance on certain cells
- To make karyotypes for chromosomal study
- To form clones of cells from one cell

Two functions of fixation of tissue

- Prevents postmortem changes (autolysis and putrefaction).
- Fixation hardens the tissue and makes it stainable.
- It decreases the shrinkage and distortion of cells.
- It preserves cell constitutes insoluble

What does acidophilic structure means {سؤال في المحاضرة}

- Basic structure in cell that has affinity to acidic stain

What does basophilic structure means

- Acidic structure in cell that has affinity to basic stain

Different steps for obtaining of tissue paraffin section{أسئلة كتاب القسم}

1. tissue sampling
2. fixation of tissue
3. Washing
4. Dehydration
5. Clearing
6. Impregnation

7. Embedding
8. Sectioning
9. Mounting
10. Staining

Four different special preparation for microscopy{أسئلة كتاب القسم}

- Filming—Smearing—Teasing—Spreading—Grinding

Plasma membrane

Complete

……is the structural and functional unit of the body

- cell

cells are bound together to form……..

- tissues

tissues are combined to form ……

- organs

……is a covering membrane which forms the outermost cover of the cytoplasm

- Cell membrane or plasma membrane

thickness of plasma membrane ranges ……to ……. nm

- 7—10

Why cell membrane cannot be distinguished (not visible) by light microscope

- because it is very thin(7-10nm) below the Lr of the L.M (240 nm)

Cell membrane by Electron Microscope appears as if it is formed of…… layers, therefore it is called …… {مهم}

- three—**trilaminar membrane**

Molecular Structure Of the Cell Membrane is formed of……. , …… and ……..

- lipids—proteins—carbohydrates

60% of cell membrane is ……., 35% …… and 5% ……..

- proteins—lipids—carbohydrates

Lipid Molecules in the Cell membrane are formed of……..and…….molecules

- Phospholipid
- Cholesterol

The Phospholipid molecule of Cell membrane is formed of……..and……

- Hydrophilic head—Hydrophobic tail

Hydrophilic Lipid of the Phospholipid Molecule forms ……but Hydrophobic Lipid of the Phospholipid Molecule forms……..

- heads—tails

Hydrophilic head of Phospholipid composed of charged lipid and is called …….

- Polar Region

Hydrophobic tail of Phospholipid composed of non charged lipid and is called …….

- Non Polar Region

…….have a stabilizing effect, helping to keep the phospholipid molecules packed together present in plasma membrane

- Cholesterol Molecules

The Large intrinsic protein molecules that extend through its entire thickness of cell membrane is called ……. proteins{مهم}

- transmembrane

The Carbohydrate molecules of the cell membrane is in the form of..... {Final2024}

- oligosaccharides

Carbohydrate attached to the lipids called……. or attached to proteins called …….

- glycolipids—glycoproteins

Carbohydrate Molecules that project to the outer surface of the cell to form ……{مهم}

- glycocalyx or cell coat

cell membrane can engulf or allow solid materials to enter into the cytoplasm by the process of ……{مهم}

- phagocytosis

solid particle after its entrance into the cytoplasm is called ……{مهم}

- phagosome

Transport Of Fluid Material By ……{مهم}

- Pinocytosis

The engulfed fluid becomes surrounded by a membrane to be changed into a……{مهم}

- Pinocytic Vesicle

Expulsion Of Residual Bodies By ……{مهم}

- Exocytosis

The cell can extrude the digested bodies outside the cell by the process of……{مهم}

- exocytosis

After the entrance of the solid or the liquid materials into the cytoplasm, they will be digested by the ……..enzymes

- lysosomal

Enumerate

The three main components of Cytoplasm{مهم}

- Cytoplasmic **Organelles**
- Cytoplasmic **Matrix**
- Cytoplasmic **Inclusions**

Four examples of Membranous Organelles {مهم} 7 {أسئلة كتاب القسم}{سؤال في المحاضرة}

- Cell membrane or plasma membrane
- Endoplasmic reticulum (Rough and smooth)
- Mitochondria
- Peroxisomes
- Golgi apparatus
- Secretory vesicles
- Lysosomes

Two examples of Non-membranous Organelles{مهم}

- Ribosomes
- Cytoskeleton (Microtubules , Filaments)
- Centrioles

Two functions of Cholesterol Molecules in cell membrane

- stabilization
- helping to keep the phospholipid molecules packed together
- increase the integration of the plasma membrane

The two types or forms of protein molecules in cell membrane

- Extrinsic or Peripheral Protein
- Intrinsic or Integral Protein

Two Functions of the Cell Membrane

- semipermeable that Control the exchange of materials between the Cell and its surroundings

- Enable Adhesion to adjacent cells
- Recognition
- Formation of Cell Immunity

Two examples of substances that cross the cell membrane by Passive Diffusion

- dissolved gases—essential ions—**oxygen**—**carbon dioxide**

Two examples of substances that cross the cell membrane by Active Transport

- Large molecules of sugars—amino acids – fatty acids

Two examples of substances that cross the cell membrane by Selective Transport

- drugs—hormones--bacteria –viruses

Two Functions Of Cell Coat Or Glycocalyx

- Cell Adhesion to adjacent cells
- Cell Recognition
- Formation Of Cell Immunity

N.B {~~24~~}

1. **Fluid mosaic nature of the PM**: At body temperature, the lipid bilayer is **fluid** allowing free movement of the integral proteins within the membrane.
2. **Unit membrane theory**: By E.M. all cellular membranes have the same **trilaminar** structure

Membranous Organelles 1

Complete

Endoplasmic Reticulum is classified into……..and…….

- Rough endoplasmic reticulum (RER) and Smooth endoplasmic reticulum (SER)

RER by L.M is not visible but their site appears as localized area of ……..

- basophilia

…….by E.M appears as tubules, vesicles or cisternae of SER attached on their surfaces ribosomes

- Rough Endoplasmic Reticulum (RER)

RER by E.M appears as tubules, vesicles or cisternae of SER and…….. attached on their surfaces

- ribosomes

…… organelle responsible for Formation of the secretory proteins and lysosomes, packing and transfer them to Golgi

- Rough Endoplasmic Reticulum (RER)

……consists of branching and anastomosing smooth tubules not studded with ribosomes

- Smooth Endoplasmic Reticulum (SER)

……organelle responsible for Lipid synthesis and Detoxification of drugs and toxins

- Smooth Endoplasmic Reticulum (SER)

…….. known as the power houses of the cell

- Mitochondria

Mitochondria produce energy in the form of ……..

- ATP

special stain for mitochondria is……..

- iron Hx

The outer mitochondrial membrane is…… and the inner mitochondrial membrane is thrown into ……

- straight—folds

the inner mitochondrial membrane is thrown into folds called ……

- cristae or crests

outer membrane is thick permeable, contains special transmembrane proteins called……. that serve like channels for transport of substances into intermembrane space

- porins

The inner mitochondrial membrane is thin, less permeable and contains…… enzymes

- respiratory chain

membranes of the crests of mitochondria show small particles known as ……

- elementary particles

The Elementary particles represent…… responsible for the generation of ……

- ATP synthetase—ATP

The mitochondrial matrix surrounded by ……

- inner mitochondrial membrane

The inner chamber is surrounded by the inner membrane and contains ……

- matrix

DNA in mitochondria is circular and inherited from ……

- mother

dense matrix granules in mitochondrial matrix represent……

- calcium particles

dense particles in mitochondrial matrix is the store of……

- calcium

Mitochondria are sometimes known as symbions as they resemble the structure of bacteria and they multiply by ……

- binary fission

……is responsible for division of mitochondria that present in the mitochondrial matrix

- DNA

Enzymes concerned with the kreb's cycle and β oxidation of fatty acids present in……..

- mitochondrial matrix

Peroxisomes called……..

- microbodies

Peroxisomes are numerous in……..

- liver cells

Peroxisomes are small spherical membranous organelles arise from ……

- RER

Peroxisomes by E.M, they are morphologically similar to…….. and appear as finely granulated with dense core called ……

- lysosomes—nucleoid

Oxidation of long chain fatty acid with production of H_2O_2 is the function of.....{Final2024}

- Peroxisomes or microbodies

……..enzyme in peroxizomes decomposes H_2O_2 by converting it to water and O_2

- catalase

Which organelle responsible for Detoxification of drugs and toxins {Final2024}

- smooth endoplasmic reticulum (SER)

Enumerate

Two Functions of RER

- Formation of the **secretory proteins**
- Formation of **lysosomes and peroxisomes**

Three Functions of SER {سؤال في المحاضرة}

- **Lipid synthesis** as adrenal cortical cells
- **Detoxification of drugs and toxins** as liver cells
- **Mineral metabolism** as cells of the lacrimal gland
- **Glycogen synthesis** as liver cells
- **Impulse transmission within the cell** as skeletal and cardiac muscle cells
- **phospholipids synthesis** for all cell membranes

Two functions of mitochondria

- Energy production
- Regulation of cytosolic calcium concentration

Two organelles involved in food oxidation {سؤال في المحاضرة}

- Mitochondria and peroxisomes

identify the EM structure of mitochondria{سؤال في المحاضرة}

- Organelle surrounded by two membranes (The outer is straight and the inner is thrown into folds called cristae or crests)

Compare the EM structure of SER and RER{سؤال في المحاضرة}

Correlate the structure of mitochondria to its function{سؤال في المحاضرة}

Membranous Organelles 2

Complete

By H&E, Golgi apparatus appear as unstained area in deeply basophilic cells called …….

- negative Golgi image

……..and……..are deeply basophilic cells by H&E showing negative Golgi image

- plasma cells—osteoblast

special stain for Golgi apparatus is……..

- silver stain

……..is formed of membrane bounded cisternae or saccules arranged in the form of parallel stacks that has a convex surface and a concave surface.

- Golgi apparatus

……..is responsible for condensation and packing of proteins

- Golgi apparatus

The convex surface of Golgi called ……..surface but the concave surface called……..

- cis or formation – trans or maturation

…….. are associated with the formation or (Cis) or (convex) surface of Golgi

- transport vesicles

…… are associated with the maturation or (trans) or (Concave) surface of Golgi

- Secretory vesicles

Secretory vesicles present throughout the cytoplasm and numerous in ……..cells

- active secretory

……formed in the endoplasmic reticulum and are processed and packed in Golgi apparatus

- Secretory vesicles

Secretory vesicles are formed in …….. and processed and packed in……..

- RER—Golgi apparatus

The materials within the secretory vesicle released to the outside of the cell by…….. as the vesicles fuse with the plasma membrane

- exocytosis

Vesicles contain digestive enzymes are called ……..

- zymogen granules

…….are spherical membranous organelles rich in hydrolytic enzyme

- Lysosomes

Lysosomes are invisible by L.M , could be demonstrated histochemically by detection of …….

- acid phophatase enzyme

……are often called 'garbage system' of the cell

- Lysosomes

Lysosomes contain hydrolytic enzymes work in…… media

- acidic

Lysosomes are budded into the cytoplasm from the…… with enzymes inside

- Golgi apparatus

Lysosomal enzymes are made in ……, which are then delivered to ……via transport vesicles.

- rough endoplasmic reticulum—Golgi apparatus

primary lysosomes are ……in electron density while secondary lysosomes are ……

- homogeneous—heterogeneous

The newly formed lysosome is known as …….. lysosome

- primary

lysosome which fuses with pinocytotic vesicle or phagosome known as…….. lysosome

- secondary

lysosome fuses with old organelle as old mitochondria forming secondary lysosome that called……

- autophagic vacuole or cytosome

Secondary lysosomes digest their contents and transformed into residual body that is either extruded outside the cell by……or remains inside the cell as……

- exocytosis—lipofescin pigment

……is A normal physiological process that deals with the destruction of cells in the body that important in turnover of destroyed cell organelles

- Autophagy

The taking into the cell of exogenous material by phagocytosis or pinocytosis and the digestion of the ingested material after fusion of the newly formed vacuole with a lysosome , this process called ……

- Heterophagy

…….refers to the killing of an entire set of cells by the breakdown of the lysosomal membrane

- Autolysis

Liquid substances enter cell by …….process

- pinocytosis

Solid substances enter cell by …….process

- phagocytosis

…….of the sperm head is a giant lysosome that ruptures and releases enzymes on the surface of the egg

- acrosomal cap

Enumerate

Three functions of Golgi bodies {سؤال في المحاضرة}

- condensation and packing of proteins
- packing hydrolytic enzymes of the primary lysosomes
- addition of carbohydrates to from glycoprotion
- sulphation of some products
- membrane traffic and recycling in the cell

Two examples of cells showing negative Golgi image

- plasma cells
- osteoblast

Two examples of enzymes in peroxizomes

- catalase—urate oxidase—fatty acid—oxidase—amino acid oxidase

Types of lysosomes

- primary lysosomes
- secondary lysosomes

Two Functions of Lysosomes

- digestion of the ingested material
- Breakdown old or damaged organelles in cell (Autophagy)
- Killing or destroying of the cell (Autolysis)
- Hydrolytic enzymes that hydrolyze proteins , lipids , carbohydrates , nucleic acids

Two important cellular in which lysosomes involved Or Role of ribosomes in biological processes{مهم} {سؤال في المحاضرة}

- Heterophagy
- Autophagy
- Autolysis
- Fertilization

Two organelles involved in formation of secretory proteins. {سؤال في المحاضرة}

- RER and Golgi apparatus

What is meant by the term "negative Golgi image" ? {سؤال في المحاضرة}

- unstained area appears in deeply basophilic cells on staining by H&E as plasma cells and osteoblasts

identify the E.M structure of Golgi bodies {سؤال في المحاضرة}

Correlate the structure of lysosomes to its function {سؤال في المحاضرة}

Non membranous organelles & cell inclusions

Complete

Ribosomes are formed from …….. and……..

- RNA—protein

Ribosomes are formed in ……

- nucleolus

Proteins are synthesized in ……..

- ribosomes

……are Small particles that Consist of two subunits and are formed of RNA and protein

- ribosomes

……..are Small particles Involved in synthesis of proteins utilized by the cell growth and differentiation

- Free ribosomes

…… are Small particles Attached to the outer surface of RER concerned with synthesis of secretory proteins exported outside the cell

- Attached ribosomes

Free ribosomes organized in the form of…….. by the aid of mRNA

- polysomes

When many ribosomes attached to the same mRNA strand to form larger complexes called……..

- polysomes

……..ribosomes synthesize cellular protein but …….. ribosomes synthesize exported protein

- Free—Attached

The wall of ……..is formed of 13 protofilaments

- microtubule

……..are formed of protein tubulin organized into thread like protofilaments

- microtubules

Displacement of macromolecules and organelles through the cytoplasm is the function of…… {Final2024}

- microtubules

Microtubules are formed of protein……. organized into thread like ……..

- tubulin—protofilaments

The Thin filaments or microfilaments are formed of…… protein

- actin

Thin filaments or microfilaments present highly organized in…….. cells

- muscle

Keratin is a type of……filaments that forming the tonofilaments in

- Intermediate—stratified epithelia

Desmin is a type of…….filaments that present in…….. cells

- Intermediate—smooth muscle

Neurofilaments is a type of…….filaments that present in……. cells

- Intermediate—nerve

Vimentin is a type of…….filaments that present in ……..

- fibroblasts

Glial filaments is a type of…….filaments that present in…….. cells

- Intermediate—glial

Thick (myosin) filaments present in…….. and…….. cells

- skeletal muscle—cardiac muscle

Thick filaments are called…… filaments

- myosin

……..appears as two dots close to the nucleus important in cell division

- centrioles

The wall of …….. formed of 9 bundles of triplet microtubules

- centrioles or basal body of cilia

…….form mitotic spindles in the microtubule organizing center during cell division

- centrioles

…….serve as the basal body of cilia and flagella

- centrioles

Enumerate

Three examples of Non-membranous organelles{أسئلة كتاب القسم}

- Ribosomes
- Centrioles
- Cytoskeleton {Microtubules—Filaments}

two types of ribosomes

- Free ribosomes
- Attached ribosomes

Constituents of cytoskeleton{سؤال في المحاضرة}

- microtubule
- microfilaments
- Intermediate filaments

Two Functions of microtubules

- Displacement of macromolecules and organelles through the cytoplasm
- Mechanical support of the cell and determine its morphology
- Form the mitotic spindles during cell division
- in important structures as centrioles, cilia and flagella

Types of Filaments {أسئلة كتاب القسم}

- Thin filaments or microfilaments or **actin** flilaments
- Intermediate filaments
- Thick filaments or **myosin** filaments

The main types of Intermediate filaments {أسئلة كتاب القسم}

- Keratin
- Desmin
- Vimentin
- Neurofilaments

- Glial filaments

Two Functions of filaments or cytoskeleton

- Mechanical support of the cell.
- Distribute the tensile force on the cell.
- Involved in muscle contraction.
- Play a role in cell division

Three Functions of centrioles {سؤال في المحاضرة}

- important in cell division
- form mitotic spindles in the microtubule organizing center
- serve as the basal body of cilia and flagella

Cell inclusions

Complete

……include non-living substances stored in the cytoplasm and not bound by membranes

- Cell inclusions

……is Stored food In liver and muscle cells

- Glycogen

Glycogen Can be localized by special stain called …… reaction

- PAS

Glycogen appears as clusters of granules either as individual dense, roughly small particles, referred to as …….. as in ……. or aggregate in two or more particles as rosette-like of larger size called ……as in …….

- β– particles—liver α- particles –muscle

……has signet ring appearance by Hx&E

- fat cells (adipocytes)

Special method for fat demonstration is…… stain of fresh sections

- Sudan black

……..is iron containing pigment that is accumulated in the cytoplasm of the phagocytes

- Hemosiderin

……..is brown granular material has accumulated as a result of the breakdown of RBC's

- Hemosiderin

……..is an Endogenous pigment in the epidermis, iris, hairs and substantia nigra

- Melanin

……..is an Endogenous pigment in the blood and liver cells

- Bilirubin

……..is a pale brown granule found in many cells especially in stable non dividing cells as …….and ……..cells

- nerve— liver — cardiac muscle

……..is an Endogenous pigment in active long lived cells

- Lipofuscin

……..is an Exogenous pigment in carrots and tomatoes

- Lipochrom or carotenoids

……..is an Exogenous pigment in lung macrophages

- Dust or carbon particles

……..is an Exogenous pigment in the gum epithelia and skin cells

- lead and tattoo marks

crystalloids of charcot- bottcher are crystals in cells of the ……..and linked with morbidity and infertility

- testis

Crystals have special function in cells of…… to maintain balance

- inner ear

Enumerate

Classification or types of Cell inclusions

- **Stored food** (as glycogen and fat)
- **Pigments** (endogenous as Hemosiderin ……. and exogenous as Lipochrom……)
- **Crystals**

Types of stored food {سؤال في المحاضرة}

- glycogen – fat

Stain used to demonstrate glycogen {سؤال في المحاضرة}

- PAS reaction

Classification or types of pigments with example

- **Endogenous** pigments as Hemosiderin—Bilirubin—Melanin—Lipofuscin
- **Exogenous** pigments as carotenoids – carbon particles

Two examples of Endogenous pigments

- Hemosiderin—Bilirubin—Melanin—Lipofuscin

Two examples of Endogenous pigments

- carotenoids – carbon particles—Minerals as lead and tattoo marks

Nucleus

Complete

……..is the control center of the cell

- Nucleus

Nucleus contains ……. which control all the structural and functional aspects of the organism.

- genes

Most human cells are ……..cells containing true nucleus

- eukaryotic

Cells may be binucleated as…… cells and may be multinucleated as …….. cells

- liver— skeletal muscle

…… nucleus contains high content of inactive condensed chromatin (Heterochromatin)

- Condensed

……..nucleus also called open-face nucleus

- Vesicular

…… nucleus contains high content of active chromatin (euchromatin)

- Vesicular or open-face

……adherent to inner surface of nuclear envelope and…….. associated to its outer surface

- chromatin—ribosomes

The outer nuclear membrane is continuous with ……and is covered with…….but the inner one is lined with ……

- rough endoplasmic reticulum—ribosomes —chromatin

……allows the exchange of special substances between the nucleus and the cytoplasm

- Nuclear pores

……..is a cylindrical channel extending from the nuclear pore into the cytoplasm

- nuclear pore complex

……materials are the basophilic particles and threads from which chromosomes are formed during cell division

- Chromatin

……is the uncoiled or less coiled chromatin that is Lightly stained

- Euchromatin

……is the More coiled chromatin that is dark stained basophilic

- Heterochromatin

……..is condensed clumps represent the inactive coiled parts of chromosomes

- Heterochromatin

……is a chromatin which is predominant in inactive cells

- Heterochromatin

……is a chromatin which is predominant in active cells

- Euchromatin

…….and ……cells are examples of cells contains euchromatin

- Liver and nerve

The DNA is organized around histones into …… that are wound into a helix to form chromatin

- nucleosomes

chromatin is wound into a supercoiled ……

- chromosomes

The complete set of genes is called …….

- genome

…….are composed of RNA and proteins

- Ribosomes

…….By E.M appears as a diffuse irregular mass not limited by a membrane composed of two areas Pars fibrosa and Pars granulosa

- nucleolus

nucleolus is composed of two areas ……and…….

- Pars fibrosa and Pars granulosa

Pars fibrosa is the central part which is formed of the nucleolar organizer chromosomes and ……

- filamentous RNA

……is the central part which is formed of the nucleolar organizer chromosomes and filamentous RNA

- Pars fibrosa

Pars granulosa represents the accumulation of …….in various stages of formation

- RNA protein

…….represents the accumulation of RNA protein in various stages of formation

- Pars granulosa

……is colloidal clear solution in which the chromatin is suspended and has fibrillar appearance

- nuclear matrix

Enumerate

Different shapes of nuclei {Final2024}

- spherical , oval , kidney shaped , spiral ,lobulated , flattened

Types of nucleus by L.M. {سؤال في المحاضرة}

- **Condensed** nucleus
- **Vesicular or open-face** nucleus

The four Components of the nucleus

- Nuclear envelope
- Chromatin
- Nucleolus
- Nuclear matrix

Types of Chromatin {سؤال في المحاضرة}

- Euchromatin
- Heterochromatin

One example of cells containing Euchromatin and One example of cells containing Heterochromatin

- Euchromatin as Liver &nerve cells
- Heterochromatin as Lymphocytes

Two types of Heterochrmatin{سؤال في المحاضرة}

- Peripheral chromatin
- Nucleolar associated chromatin
- Barr body
- Scattered clumps

Components or parts of nucleotide in DNA

- Deoxyribose sugar
- Nitrogenous base
- Phosphategroup

The four nitrogenous bases present in DNA

- Adenine—thymidine – cytosine—Guanine

The four nitrogenous bases present in RNA

- Adenine—Uracil – cytosine—Guanine

Components of Chromatin structure

- DNA double helix
- Small amount of RNA
- Histones
- Non- histones

Two areas in nucleolus

- Pars fibrosa
- Pars granulosa

function of nucleolus

- site of transcription of ribosomal RNA that is concerned with protein synthesis

Why the nucleus is basophilic

- because it is rich in nucleic acids (DNA & RNA)

N.B nuclear envelope is formed of 2 membranes (8 nm thick each) with a 25 nm wide space in between

Structure of chromosome & karyotyping

Complete

Types of cells are ……cells which are present in the whole tissues of the body and Gametes which are the ……and…….

- Somatic—sperms and ova

Any cell in our body is somatic except ……..and……..

- sperms and ova

Each somatic cell nucleus contains ……chromosomes

- 46

Each gamete (ovum or sperm) contains……. chromosomes

- 23

22 pairs of identical chromosomes known as …….

- autosomes

……..is the study of the number, type and morphology of chromosomes of an individual

- Karyotyping

Karyotyping is done by treating a number of dividing cells with……. to arrest them in…… stage

- cholchicine—**metaphase**

Each pair is identical in shape, size and structure and is called…… chromosomes

- homologous

Each chromosome is formed of two chromatids which are connected together at one point known as ……

- centromere

The centromere appears as light- stained region except at its center, a dense small granule is present called …….

- spherule

……..is a disc shaped structure present in the centromere acts as microtubule organizing center during cell division

- Kinetochore

microtubule organizing center in chromosome during cell division is……..

- Kinetochore

microtubule organizing center in cytoplasm during cell division is……..

- Centrioles

in ……..chromosome,the centromere is located centrally

- Metacentric

in …….. chromosome , the centromere is present between the centers of the chromatids and their ends

Or in …….. chromosome , the centromere is present near the center

- Submetacentric

in …….. chromosome, the centromere is present near the terminal ends of the chromatids

- Acrocentric

in …….. chromosome , the centromere is present at the terminal ends of the chromatids

- Telocentric

In…….. Chromosome, the Centromere is present in terminal end of chromatin {Summar2024}

- Telocentric

……..is a chromosomal abnormality in which the cell contains more than 2 haploid number of chromosomes e.g. cancer cells

- polyploidy

the trisomy 21 results in ……

- Down syndrome

Down syndrome is ……..type of chromosomal abnormality

- Anuploidy

……..is a chromosomal abnormality occurs loss of an arm of a chromosome

- Deletion

…….is a chromosomal abnormality occurs addition of extra piece to a chromosome

- Duplication

…….is a chromosomal abnormality occurs when a chromosome breaks and re-attach in a reverse position

- Inversion

…….is a chromosomal abnormality occurs when a chromosome divides in a transverse manner instead of longitudinally

- Isochromosome

Types of Chromosomes{سؤال في المحاضرة} = Types of Chromosomes According to centromere,

- Metacentric
- Submetacentric
- Acrocentric
- Telocentric

three types of chromosomal abnormality {سؤال في المحاضرة}

- polyploidy
- Anuploidy
- Deletion
- Duplication
- Inversion
- Isochromosome

Two types of chromosomal abnormality in the number

- polyploidy—Anuploidy

What is meant by Anuploidy {سؤال في المحاضرة}

- the trisomy 21 results in Down syndrome

Two types of chromosomal abnormality in the structure

- Deletion—Duplication—Inversion—Isochromosome

Cell cycle &cell death

Complete

……..is the time from the end of mitosis to the end of the following mitosis

- cell cycle

cell cycle is formed of 2 phases, ……and ……..

- interphase—mitosis

……is the interval between 2 consecutive cell divisions during which the cell performs its specialized functions

- Interphase

Interphase divided into 3 phases are …….. , ……. and……

- G1 phase—S phase—G2 phase

…….. phase of cell cycle during which the chromosomes are formed of single chromatids

- G1

G0 stage of cell cycle is called…… stage

- resting

…….and…….. are cells enters G0 stage (resting stage) for its life span

- nerve cells and cardiac muscle

DNA replication occurs in which phase? {Summar2024}

- Synthesis (S) phase

……..phase during which DNA and histones replicate and the chromosomes are formed of double chromatids

- Synthesis (S)

…… phase during which, the cell synthesize a great amount of cytoplasmic material

- G2

……..is a type of cell division occurs in somatic cells

- Mitosis

in …… stage of mitosis the two centrioles migrate to the opposite side and Microtubules arise radially between the them

- prophase

in …… stage of mitosis chromosomes continue to shorten and thicken and align in the equatorial plane of the cell

- Metaphase

each chromosome splits into two chromatids at the centromere in ……. stage of mitosis

- Anaphase

contraction of microtubules to separate sister chromatids occurs in ……. stage of mitosis

- Anaphase

in …… stage of mitosis constriction occurs in the equatorial plane which deepens leading to cleavage of the cell into 2 daughter cells

- Telophase

nuclear envelope is reassembled in …… stage of mitosis

- Telophase

……is increase in all mass through increase in their number by cell division (mitosis)

- Growth

……..is the process by which new features and new functions are achieved.

- Differentiation

……..is the process by which growth and differentiation are kept in harmony resulting in the formation of normal organism

- Integration

……..is called programmed cell death

- Apoptosis

Alternative term for Apoptosis is…… {أسئلة كتاب القسم}

- programmed cell death

……is the cell death due to exogenous factors such as injury, toxins or hypoxia

- Necrosis

in Necrosis, cell bursts and scattered fragments are removed by ……

- phagocytosis

in apoptosis , apoptotic bodies are phagocytized by ……..

- neighboring cells

……is a type of cell death that usually involve cell swelling and destruction of organelles

- Necrosis

……is a type of cell death that usually involve cell shrinkage but the organelles remain normal or intact

- Apoptosis

……..is Cell death due to endogenous factor which is activation of certain genes.

- Apoptosis

What type of cell death caused by endogenous factors? {سؤال في المحاضرة}

- Apoptosis

in Apoptosis, cell fragmented into membrane-bounded globules known as ……that are phagocytized by neighboring cells

- apoptotic bodies

Enumerate

The stages of the cell cycle{أسئلة كتاب القسم} لازم الترتيب

1. Gap l (G1) phase
2. Synthesis (S) phase
3. Gap 2 (G2) phase
4. Mitosis

Enumerate in order the 4 stages of the cell cycle {سؤال في المحاضرة} لازم الترتيب

1. Gap l (G1) phase
2. Synthesis (S) phase
3. Gap 2 (G2) phase
4. Mitosis

The Three phases of interphase

- Gap l (G1) phase
- Synthesis (S) phase
- Gap 2 (G2) phase

The phases of mitosis{أسئلة كتاب القسم}

1. prophase
2. Metaphase
3. Anaphase
4. Telophase

Two examples of cells enters G0 stage (resting stage)

- nerve cells

- cardiac muscle

three aspects or events of Cell development

- Growth
- Differentiation
- Integration

types of cell death{أسئلة كتاب القسم}

- Necrosis
- Apoptosis (programmed cell death)

Two events occurs in Necrosis or The main cellular changes occur in necrosis{سؤال في المحاضرة}

- cell swelling
- organelles are destroyed
- chromatin is clumped and the cytoplasm becomes weakly stained
- cell bursts are removed by phagocytosis

Two events occurs in apoptosis or The main cellular changes occur in apoptosis {سؤال في المحاضرة}

- cell shrinkage
- condensation and fragmentation of DNA
- organelles remain normal
- cell fragmented into membrane-bounded globules known as apoptotic bodies
- apoptotic bodies are phagocytized by neighboring cells

Compare between apoptosis and necrosis(3 points) {سؤال في المحاضرة}

- in apoptosis cell shrinkage occurs but in Necrosis cell swelling occurs
- in apoptosis organelles remain normal but in Necrosis organelles are destroyed
- in apoptosis apoptotic bodies are phagocytized by neighboring cells but in Necrosis cell bursts are removed by phagocytosis

define cell differentiation{سؤال في المحاضرة}

- process by which new features and new functions are achieved

Stem cells and tissue renewal

Complete

……..are Undifferentiated Cell has the ability to continuously divide and differentiate (develop) into various other kind(s) of cells/tissues

- Stem cells

cells of one type of tissue can give rise to cells of different type of tissue , this called……

- plasticity

……stem cells can develop into a new individual (embryo and placenta)

- Totipotent

Fertilized egg, Cells from early embryos (1-3 days) is an example of ……..stem cells

- Totipotent

…….. is an example of totipotent stem cells

- Fertilized egg

……. stem cells can form the three germ layers of the embryo:ectoderm, mesoderm, endoderm

- Pluripotent

……stem cells can Form embryo without placenta

- Pluripotent

cells of inner cell mass of the blastocyst is an example of ……..stem cells

- Pluripotent

……is an example of Pluripotent stem cells

- cells of inner cell mass of the blastocyst

……stem cells can generate different cell types for the specific tissue or organ in which they live

- Multipotent

blood forming or hematopoietic stem cells in the bone marrow is an example of ……..stem cells

- Multipotent

…….is an example of Multipotent stem cells

- hematopoietic stem cells

……stem cells can differentiate into a few cells

- Oligopotent

lymphoid or myeloid blood forming cells in bone marrow is an example of ……..stem cells

- Oligopotent

…….is an example of Oligopotent stem cells

- lymphoid cells or myeloid cells

……stem cells can differentiate only Into two types of cells

- Bipotent

Hepatic progenitor cells is an example of ……..stem cells

- Bipotent

…….is an example of Bipotent stem cells

- Hepatic progenitor cells

Hepatic progenitor cells are Bipotent stem cells can differentiate Into……. and …….. cells

- liver and biliary cells

……stem cells only produce cells of their own type

- Unipotent

Epidermal stem cells is an example of ……..stem cells

- Unipotent

……is an example of Unipotent stem cells

- Epidermal stem cells and Muscle stem cell

Embryonic stem cells derived from……..

- inner cell mass of the blastocyst

markers for identification of stem cell are……..and……..

- CD45 and CD117

Theoretically and under research stem cell therapies could be applied in ……..

- regenerative medicine

Currently, the only stem cell-based treatment that is routinely reviewed and approved by Food and Drug Administration (FDA) is ……

- hematopoietic stem cell transplantation or bone marrow transplantation

Skin epidermis renewal occurs in…….. weeks and Intestinal epithelium renewal occurs in …….. days

- three—four

Enumerate

Two Characteristics of Stem cells {Summar2024}

- Undifferentiated cells
- Capable of proliferation and self-renewal
- Capable of Differentiation
- Capable of plasticity

Types of stem cell according to differentiation capability (capacity or potentiality) and give an example for each one {سؤال في المحاضرة}

- **Totipotent** as Fertilized egg
- **Pluripotent** as cells of inner cell mass of the blastocyst (5 to 7 days)
- **Multipotent** as hematopoietic stem cells in the bone marrow
- **Oligopotent** as lymphoid or myeloid blood forming cells in bone marrow
- **Bipotent** as Hepatic progenitor cells
- **Unipotent** as Epidermal stem cells and Muscle stem cell

Types of stem cell according to its source (origin) {سؤال في المحاضرة}

- Embryonic stem cells
- Adult stem cells
- Induced pluripotent stem cells

Two sites of Adult stem cells

- Bone Marrow
- Adipose tissue-derived stem cells
- Umbilical cord stem cells
- Human amniotic fluid
- Tissue-specific stem cells as in skin , liver

What is the Applications of Stem Cell in medicine

- spinal cord injuries
- type 1 diabetes
- Parkinson's disease,
- Alzheimer's disease
- myocardial infarction
- burns
- cancer
- osteoarthritis

what is meant by tissue bank? {سؤال في المحاضرة}

- stem cells can be stored and used in therapies to prevent the future life-threatening diseases as umbilical cord stem cells and Stem cells of human exfoliated deciduous teeth

What is the physiological role of stem cells

- in Cell renewal for repair of damaged tissue and also maintain the pool of undifferentiated stem cells

Define stem cells{سؤال في المحاضرة} {أسئلة كتاب القسم}

- Undifferentiated Cell has the ability to continuously divide and differentiate into various other kinds of cells

Define Pluripotent stem cells{أسئلة كتاب القسم}

- cells can form the three germ layers of the embryo: ectoderm, mesoderm, endoderm as inner cell mass of blastocyst

Define Totipotent stem cells{أسئلة كتاب القسم}

- cell can develop into a new individual (embryo and placenta) as Fertilized egg

Define Multipotent stem cells{أسئلة كتاب القسم}

- can generate different cell types for the specific tissue or organ in which they live as hematopoietic stem cells

Histological structure of the neurons

Complete

…… is the structural and functional unit of nervous tissue

- Neuron

nerve cell has different shape with pale acidophilic cytoplasm with deep basophilic masses of ……..

- Nissl bodies

Type of nucleus in nerve cell (Neuron) is…….. nucleus

- vesicular

The processes attached to cell body are ……..and…….

- axon and dendrite(s)

staining of nerve cell body by ……shows neurofibrils that appears brown or black

- silver stain

…… Is yellow brown color waste material stored inside the lysosomes in active long lived cells

- Lipofuscin pigment

…….pigment present numerous in older nerve cells

- lipofuscin

……is called nerve fiber

- Axon

point of axon origin is called …….. that is free of ……

- axon hillock—Nissl bodies

Myelin sheath wrapped spirally around the …….

- axon

……cell responsible for formation of Myelin sheath in peripheral nervous system

- Schwann

Myelination of PNS occur by …….. {سؤال فى المحاضرة}

- Schwann cell

myelin sheath is composed of…… material of Schwann cell

- lipoprotein

Myelin sheath is deficient (absent) at certain points along the axon known as ……

- nodes of Ranvier

Myelin sheath is interrupted at periodic intervals by ……

- nodes of Ranvier

Myelin sheath can be demonstrated using……. stains

- osmic

Enumerate

E.M structure of Perikaryon (cell body) {سؤال في المحاضرة}

- **euchromatic** nucleus with distinct nucleolus
- **abundant RER** and **polyribosomes** (Nissl bodies), **neurofilaments, microtubules**, **Golgi apparatus**, **mitochondria**, **lysosomes** and **lipofuscin pigment**

Types of neural processes

- Dendrite
- Axon

Types of axons{سؤال في المحاضرة}

- myelinated axons
- unmyelinated axons

Epithelium

Complete

The epithelium of the mouth are of ----------origin {سؤال في المحاضرة}

- ectodermal

The cells lie on……{سؤال في المحاضرة}

- **basal lamina** or basement membrane

Nutrition of epithelium occurs by…… {سؤال في المحاضرة}

- **diffusion** from the underlying CT.

Zonula occludens is also called……{سؤال في المحاضرة}

- tight junction

function of nexus is …… {سؤال في المحاضرة}

- **rapid transport** of impulse from one cell to another

The special stain that detect basal lamina is ----------{سؤال في المحاضرة}

- PAS

The epithelium of the Skin are of ……origin

- ectodermal

The epithelium of the nose and anus are of ……origin

- ectodermal

The epithelium of the respiratory and digestive tracts are of ……origin

- endodermal

The epithelium lining blood vessels (Endothelium) are of ……origin

- mesodermal

Epithelial cells lie on a fibrous lamina called …….

- **basal lamina** or basement membrane

The epithelium is avascular and its nutrition occurs by……. from the underlying CT. blood vessels

- diffusion

epithelial cells that are composed of one cell layer called……..shows the property of epithelial ……

- simple epithelium—polarity

transformation of one type of epithelial tissue into another type called ……

- metaplasia

Basal lamina was called ……..

- basement membrane

The special stain of Basal lamina or basement membrane is……

- PAS

Basal lamina or basement membrane By E.M composed of two zones …….. and……

- lamina lucida and lamina densa

…….is traversed by very fine strands of fibers that extends from the cell membrane to the lamina densa.

- Lamina lucida

……..is formed of a network of randomly oriented interwoven filaments that are formed of collagen fibers, proteoglycans and laminin.

- lamina densa

The basal lamina is attached to the underlying C.T. by ……..

- anchoring fibrils

The cell junction which seals the intercellular spaces is……

- Zonula occludens or tight junction

…….is the most apical of the junctional complex seen between epithelial cells

- Zonula occludens or tight junction

in ……., the intercellular space filled with adhesive material mainly formed of glycoprotein

- Zonula adherens or intermediate junction

in ……. ,The adjacent cytoplasm is dense and contains tonofilaments and is called terminal web

- Zonula adherens or intermediate junction

The adjacent cytoplasm in Zonula adherens is dense and contains tonofilaments and is called……

- terminal web

Macula adherens is called…….

- desmosome

…….is a type of cell junctions is found between cells exposed to mechanical stress as cells of…….

- deomosome—epidermis

…….is the strongest junction between cells

- deomosome

in desmosome , Numerous intermediate filaments come out of the PM of the two cells and loop in a dense discoid material found midway between the cells which is called …….

- attachment plaque

in desmosome , the adjacent cytoplasm on both side of the junction is rich in ……

- tonofilaments

……. is a type of cell junctions allow free movement of ions from one cell to other

- Nexus or gap junction

Gap junction is a transmembrane integral proteins of the facing PM form a continuous ionic channel known as ……allow free movement of ions from one cell to other

- connexons

…… is a type of cell junctions represent regions of low electrical resistance

- Nexus or gap junction

…… is a type of cell junctions helps to transmit impulse rapidly from one cell to another

- Nexus or gap junction

Enumerate

Three General characters of epithelium {اسئلة كتاب القسم}

- **avascular**
- The cells **lie on basal lamina**
- **close cell-to-cell contacts**
- Epithelial cells are traversed by **numerous nerve endings**
- **simple epithelium shows epithelial polarity**

Types of Intercellular junctions

- **tight junction** or Zonula occludens
- **Zonula adherens** or intermediate junction
- **deomosome** or Macula adherens
- **gap junction ,** Nexus, or electric coupling

Apical cell specializations and covering &lining epithelium

Complete

The function of microvelli is ……..{سؤال في المحاضرة}

- **increase absorptive surface area**

…… is an apical modification Increases absorptive surface area {Summar2024}

- Microvilli

Caveoles is common in certain cells as ……..{سؤال في المحاضرة}

- **endothelium** of blood vessels

The function of stratified squamous epithelium is ……..{سؤال في المحاضرة}

- protection

Skin epidermis is formed of what type of epithelium ……..{سؤال في المحاضرة}

- keratinized stratified squamous epithelium

Cilia is formed of three parts …….. , …….. and …….. {سؤال في المحاضرة}

- Shaft , Basal body and Rootlet

Apical cell modifications or specializations include…… , …….. , ……. and ……..

- Cilia—Microvilli—Caveoles—Cuticle

……motile hair like processes extends from the free surface of some cells

- Cilia

……part of cilia that projecting from the surface of the cell and surrounded by the PM

- Shaft or axoneme

Shaft of the cilia also called……

- Axoneme

……part of cilia is formed of 9 bundles of triblet microtubules but …… is formed of 9 peripheral bundles of doublet microtubules with 2 central singlet MT

- Basal body—Shaft

……part of cilia organized from the centriole and have the same structure

- Basal body

……..part of cilia is minute fibers which anchor the basal body to the cytoplasm

- Rootlet

The function of cilia is…….

- **Movement** in one direction

slender processes project from the surface of absorptive cells called……

- Microvilli

……forms brush border

- Microvilli

……. by L.M give the absorptive surface brush like appearance

- Microvilli

…….. are small invaginations on the cell membrane resulted from the process of endocytosis and exocytosis

- Caveoles

…… is rich protein substance present in cells lining the urinary passages {Final2024}

- cuticle

Cuticle is an apical cell modification is present in epithelial cells of…… {Final2024}

- urinary passages or urinary bladder

The function of cuticle is ……

- protection

an epithelium made of a single layer of cells called……..

- Simple epithelium

an epithelium made of multiple layers of cells called……..

- Stratified epithelium

Thin polygonal cells with flattened nuclei present in ……epithelium

- Simple squamous

Lung alveoli lined by……epithelium {Summar2024}

- Simple squamous

parietal layer of Bowman's capsule of the kidney lined by ……. epithelium

- Simple squamous

blood vessels lined by ……epithelium

- Simple squamous

follicles of the thyroid gland lined by…… epithelium

- Simple cuboidal

gall bladder lined by…… epithelium

- Simple columnar epithelium

in ……..epithelium , All the cells lie on the basal lamina but only a portion of them reach the surface

- Pseudo stratified columnar epithelium

keratinized stratified squamous epithelium present in ……..

- Skin

oral cavity lined by ……..epithelium

- Stratified squamous

conjunctiva lined by ……..epithelium

- Stratified squamous

vagina lined by ……..epithelium

- Stratified squamous

uroepithelium is ……..epithelium

- Transitional

urinary bladder lined by ……..epithelium

- Transitional

Transitional epithelium found in …….

- urinary passages (urinary bladder)

dome shaped cells found in ……epithelium

- Transitional

Enumerate

Three examples of Apical cell modifications or specializations

- Cilia—Microvilli—Caveoles—Cuticle

The three parts of cilia by E.M

- Shaft—Basal body –Rootlet

The function of cilia

- **Movement** in one direction to remove fluids and small particles

The function of microvilli

- **increase the surface area for absorption**

One site have Cilia

- **Respiratory** epithelium

Two site have microvilli

- **intestinal** cells
- **kidney** tubule lining cells

One site have caveoles

- Capillary **endothelium**

Classification or Types of epithelium according to number of cell layers

- **Simple** epithelium
- **Stratified** epithelium

Types of Simple epithelium

- **Simple squamous** epithelium
- **Simple cuboidal** epithelium
- **Simple columnar** epithelium
- **Pseudo stratified** epithelium

Types of Stratified epithelium

- **Stratified squamous** epithelium
- **Stratified columnar** epithelium.
- **Transitional** epithelium

Two sites lined by Simple squamous epithelium

- **Lung alveoli**
- **Endothelium** lining blood vessels
- **Mesothelium** lining body cavities
- parietal layer of Bowman's capsule of the kidney
- thin segment of loop of Henle

Two sites of Simple cuboidal epithelium

- kidney tubules
- thyroid gland

Two sites of Simple columnar epithelium

- stomach
- small intestines
- large intestines
- gall bladder

Two sites of stratified squamous epithelium {Final2024}

- **skin**—**oral cavity**—**epiglottis**—**oesophagus**—**conjunctiva**—cornea—**vagina**—distal portion of the urethra

Secretory epithelium & glands

Complete

Salivary gland is considered as ……..gland {سؤال في المحاضرة}

- Compound tubulo-acinar

an example of uni cellular gland is ……..{سؤال في المحاضرة}

- goblet cell

give an example of coiled tubular gland ……{سؤال في المحاضرة}

- sweat glands

give an example of compound acinar gland ……..{سؤال في المحاضرة}

- pancreas

give an example of steroid secreting cell……..{سؤال في المحاضرة}

- endocrine cells of the **ovary**, **testes** and **adrenal glands**

Glands classified according to mode of secretion into …….. , ……. and ……..

- Merocrine—Apocrine—Holocrine

in…… glands ,the product is excreted by exocytosis without discontinuity in the plasma membrane

- Merocrine

in Merocrine glands, the product is excreted by ……..

- exocytosis

…… is an example of Merocrine glands

- sweat gland

in……glands , lipid droplets are surrounded by a thin film of cytoplasm and a portion of the cell membrane project from the apex of the cell

- Apocrine

……. is an example of Apocrine glands

- mammary gland

in……. glands , cells break down with an outpouring of cytoplasm and accumulated lipid-rich secretion

- Holocrine

……. is an example of Holocrine glands

- sebaceous gland

…… is an example of Simple tubular glands

- glands of intestinal mucosa

…… is an example of Simple coiled tubular glands

- sweat glands

…… is an example of Simple branched acinar glands

- sebaceous glands

…… is an example of Compound acinar gland

- pancreas

An example of compound tubular gland are…… and ……. {أسئلة كتاب القسم}

- gastric glands and bulbourethral glands

…… is an example of Compound tubulo-acinar glands

- major salivary glands

…… and ……are examples of Compound saccular glands

- mammary glands and prostate

…… glands synthesize signaling molecules (hormones) that are carried in the blood to act on distant target organs

- Endocrine

A common feature of endocrine glands is their abundance of …… {أسئلة كتاب القسم}

- vascularity

…… is an example of Polypeptide secreting endocrine glands

- beta cells of pancreas

Enumerate

Classification of glands with example{أسئلة كتاب القسم}

- **Exocrine** Glands as salivary glands , ……
- **Endocrine** Glands as pituitary gland ,…..
- **Mixed** Glands as pancreas ,…..

Classification of Exocrine Glands According to the number of cells with example

{أسئلة كتاب القسم}

- **Unicellular** glands as **goblet cell**
- **Multicellular** glands as salivary glands ,sweat glands , parotid gland , pancreas , ……

Classification of Exocrine Glands According to the nature of secretion with example

{أسئلة كتاب القسم}

- **Serous** glands e.g. parotid gland.
- **Mucous** glands e.g. oesophageal glands.
- **Seromucous** glands e.g. submandibular and sublingual.
- **Glands of special nature** e.g.mammary glands, ceruminous glands, sebaceous glands, lacrimal glands, and sweat glands.

Classification of Exocrine Glands According to the mode of secretion{أسئلة كتاب القسم}

- **Merocrine** glands as **sweat** glands
- **Apocrine** glands as **mammary** gland
- **Holocrine** glands as **sebaceous** gland

Depending on the basis of chemical nature of their hormones, endocrine glands are:

{أسئلة كتاب القسم}

- **Protein secreting** endocrine glands.
- **Polypeptide secreting** endocrine glands.
- **Glycoprotein secreting** endocrine glands.
- **Steroid secreting** endocrine glands.

Connective tissue: Cells

Complete

Types of Fibers in C.T are …….. , …….. and ……..

- collagen—elastic fibers—reticular fibers

……is the most widespread of the connective tissues

- Loose connective tissue

Fixed C.T cells include …… , …… , …… and …… {اسئلة كتاب القسم}

- fibroblasts, fat cells, mesenchynal cells and pericytes

Free C.T cells include …… , …… , …… and …… {اسئلة كتاب القسم}

- mast cells, plasma cells, macrophages, Foreign Body Giant Cells and leukocytes

……..are branched cells represent the origin of all types of connective tissue cells

- Mesenchymal cells

……are the dominant cell type of connective tissue

- Fibroblasts

…….are the largest of the connective tissue cells

- Mast Cells

……. cells are the origin of Fibroblast , fat cells ,

- Mesenchymal

Old fibroblasts are called …….

- fibrocytes

…….. are cells responsible for secretion of C.T fibers and Repair of wounds

- fibroblasts

……are fixed C.T cells that synthesize and store large quantities of lipid

- Fat cells or adipocytes

By H&E stain,…… appear as signet ring appearance

- fat cells or adipocytes

by H & E stain, fat cells appear as empty spaces which have…….. appearance

- signet ring

Pericytes are present associated with the endothelium of …….. and ……..

- blood **capillaries—metarterioles**

Pericytes are present around …….. and ……

- blood **capillaries—metarterioles**

…… is the origin of Macrophage (Histiocyte)

- monocyte

The monocytes migrate into the connective tissue, where they mature and are called ……

- macrophages

In the liver , macrophages are named ……

- kupffer cells

In the central nervous system, macrophages are named ……

- microglia

in bone tissue, macrophages are named ……

- osteoclast

……Originate when a foreign material is too large to be phagocytosed by the macrophages

- Foreign Body Giant Cell

……formed when fusion of two or more macrophages or monocytes

- Foreign Body Giant Cell

…….. is the origin of plasma cell

- B lymphocyte

Origin of plasma cell ? {Summar2024}

- B-lymphocyte

……arise in the connective tissues by further differentiation of B lymphocytes

- Plasma cells

……are rounded cells with eccentric nucleus with negative Golgi image

- Plasma cells

The function of plasma cells is…… {اسئلة كتاب القسم}

- Secretion of immunoglobulin (antibodies)

……are cells secrete immunoglobulin (antibodies)

- plasma cells

The neucleus of …… has characteristic clock face or cart wheel appearance

- plasma cells

…… are cells have intensely basophilic granules that contain histamine and heparin

- Mast Cells

The function of mast cells is ……{اسئلة كتاب القسم}

- secretion of histamine and heparin

Mast cells have deep basophilic granules that contains …… and ……..

- histamine and heparin

Function of Mast Cells is ……

- Secretion of heparine and histamine

Enumerate

Mention fixed C.T CELLS {سؤال في المحاضرة}

- fibroblasts, fat cells, mesenchynal cells and pericytes

MENTION FREE C.T CELLS {سؤال في المحاضرة}

- mast cells, plasma cells, macrophages and leukocytes

Describe one of the fixed CT cells {سؤال في المحاضرة}

مثلا نوصف ال

Fibroblast

- **By L.M**

Branched fusiform or flattened stellate cells with abundant basophilic cytoplasm and oval pale nucleus with prominent Nucleolus

- **By E.M**

contains Numerous RER , Well developed Golgi , mitochondria

Describe one of the free CT cells{سؤال في المحاضرة}

مثلا نوصف ال

Plasma cell

- **By L.M**

rounded with eccentric nucleus and deeply basophilic cytoplasm with **negative Golgi image**

- **By E.M**

contains Numerous RER , Well developed Golgi , mitochondria , and **clock face or cart wheel nucleus**

Classification of Connective tissue cells with example

- **Fixed cells** as fibroblasts, fat cells, mesenchynal cells and pericytes

- **Free cells** as mast cells, plasma cells, macrophages and leukocytes

Three examples of Fixed cells in C.T

- fibroblasts, fat cells, mesenchynal cells and pericytes

Three examples of Free cells in C.T

- mast cells, plasma cells, macrophages, Foreign Body Giant Cells and leukocytes

Two functions of fibroblasts

- Secretion of C.T fibers
- Secretion of amorphous material
- Repair of wound

Function of macrophages {اسئلة كتاب القسم}

- Phagocytosis
- Antigen presenting cell
- Secretion of cytokines as interleukin 1 to attract lymphocyte
- chemotactic for neutrophils

Connective tissue: fibers

Complete

…….fibers are elongated or wavy bundles which are formed of fibrils and have longitudinal striations

- Collagen

……gives gelatin on boiling

- collagen

……..and …… fibers have periodicity but ……fibers have no periodicity

- collagen –reticular—elastic

Each collagen fiber is composed of parallel ……

- fibrils

Each fibril is formed of parallel aggregates of ……

- micro-fibrils (tropcollagen)

Collagen type I present in …… , …… , …… and ……

- dermis, bone, tendon and capsule of organs

bone contains collagen type……

- 1

tendon contains collagen type……

- 1

capsule of organs contains collagen type……

- 1

dermis contains collagen type……

- 1

Collagen type II present in ……

- hyaline and elastic cartilage

hyaline and elastic cartilage contains collagen type……

- II (2)

Collagen type III called ……

- reticular fibers

reticular fibers is collagen type ……

- III (3)

Collagen type IV present in …….

- basal lamina of epithelia

Collagen type …….present in basal lamina of epithelium {Final2024}

- IV (4)

Collagen type……. is associated with the external lamina of smooth and striated muscle fibers

- V (5)

Elastic or yellow fibers are mainly composed of the protein ……. while Collagenous or White fibers are Composed of the protein…….

- elastin—collagen

White fibers is also called ……. fibers and yellow fibers also called ……. fibers

- Collagen—Elastic

…….are long, thin, branched fibers which have no longitudinal striations

- Elastic fibers

…….are well demonstrated with orcein stain where they take a dark brown color

- Elastic fibers

……. is the special stain for Elastic (yellow) fibers but ……. is the special stain for Reticular fibers

- orcein stain—sliver stains

There is no periodicity in……. fibers

- elastic

Function of collagen fibers is ……

- provide **strength** of the tissue

Function of elastic fibers is ……

- provide **elasticity and strength** to the tissue

Function of reticular fibers is ……

- provide **framework of the organs**

…….fibers are not visible by H &E

- Reticular

Enumerate

Enumerate types of CT fibers {سؤال في المحاضرة}

- **collagen**—**elastic** fibers—**reticular** fibers

How to demonstrate elastic fibers {سؤال في المحاضرة}

- by orcein stain

How to demonstrate reticular fibers {سؤال في المحاضرة}

- by sliver stain

Describe the structure of collagen fibers {سؤال في المحاضرة}

describe the structure of elastic fibers{سؤال في المحاضرة}

Two Functions of the C.T

- **supporting and binding other structural elements together**
- **forms the capsules of organs**
- **has a role in defense mechanism and immunity**
- **contains blood vessels**

- **nutrients and waste products diffuse** through it

Two Sites of Loose (areolar) C.T

- in sites where relatively little resistance to stress is required
- fills spaces between groups of muscle cells
- Supports epithelial tissue
- Forms a layer that sheathes the lymphatic and blood vessels
- found in the papillary layer of the dermis
- serosal linings of peritoneal and pleural cavities
- In glands

Classification of Connective Tissue According to the amount of the intercellular substance and the type of cells

- **Embryonic Connective tissue** including Mesenchymal and Mucous C.T
- **Connective tissue proper** including Loose and Dense C.T
- **Connective tissue with special properties** including Adipose tissue , blood ,Hematopoietic tissue ,Cartilage and Bone

Types of Embryonic Connective tissue

- **Mesenchymal** connective tissue.
- **Mucous** connective tissue

Types of Connective tissue proper

- **Loose** (areolar) C.T
- **Dense** regular and irregular C.T

Three types or examples of Connective tissue with special properties

- Adipose tissue
- Blood
- Hematopoietic (Lymphatic and myeloid) tissue.
- Cartilage
- Bone

Two sites of collagen fibers

- **Type I** in **dermis, bone, tendon and capsule** of organs
- **Type II** in **hyaline and elastic cartilage**
- **Type III** forms the **reticular fibers**
- **Type IV** In the **basal lamina** of epithelia

Two sites of elastic fibers

- Connective tissue through the body
- connective tissue in the **alveolar** septa
- **large blood vessels** as the **aorta**

C.T. proper: Ground substance &Types

Complete

The amorphous intercellular substance contains…… and……{اسئلة كتاب القسم}

- macromolecules—Soluble proteins—Water

……provides the framework of many parenchymatous organs{اسئلة كتاب القسم}

- Reticular fibers

The function of the amorphous material is……..{اسئلة كتاب القسم}

- Transport of nutrients and oxygen from capillaries to tissues by diffusion

Dense connective tissue is either……or……{اسئلة كتاب القسم}

- Dense white fibrous C.T or Dense elastic (yellow) C.T

……is the most widespread of the connective tissues

- Loose connective tissue

The Amorphous material is Watery and soft as in ……

- loose C.T

The Amorphous material is Firm and flexible as in …….

- cartilage

The Amorphous material is Hard and calcified as in …….

- bone

The Amorphous material is Liquid as in …….

- blood

The macromolecules of Ground Substance (Amorphous Matrix) are …… , …….. and ……

- Glycosaminoglycan—Proteoglycans—Adhesive glycoproteins

……are long un-branched polymers of repeating disaccharides of two groups

- Glycosaminoglycan

None sulfated mucopolysaccharides(Glycosaminoglycans) in the form of ……..

- hyaluronic acid

A component of amorphous material formed of protein cores to which various glycosaminoglycan are covalently linked is..... {Final2024}

- proteoglycans

……..are protein cores to which various glycosaminoglycans are covalently linked

- Proteoglycans

……attaching various components of the extracellular matrix to one another and to the cell membrane

- Adhesive glycoproteins

The increase in the amount of tissue fluid than normal results in a condition called

- edema

Dense regular white fibrous C.T is present in …… {مهم}

- tendon

Dense irregular white fibrous C.T is present in ……{مهم}

- sclera of eye

Dense regular elastic C. T is present in ……{مهم}

- ligamentum nuchae

Dense irregular elastic C. T is present in ……{مهم}

- media of the aorta

site of dense irregular elastic C. T ? {Summar2024}

- media of the aorta

…… adipose tissue is composed of cells that contain one large central droplet of yellow fat in their cytoplasm

- Unilocular or white

…….. adipose tissue is composed of cells that contain numerous lipid droplets and abundant brown mitochondria

- Multilocular or brown

……..Connective Tissue provides the framework of the myeloid and lymphoid organs

- Reticular

……..are fibroblasts specialized for secreting the constituents of reticular fibers

- Reticular cells

…… Fibers can be demonstrated well with sliver stains where they take a black color {Summar2024}

- Reticular

…… are cells that secrete reticular fibers in myeloid (bone marrow) and lymphoid (lymph nodes, spleen) C.T , but in other C.T ……are the cells that secrete the reticular fibers

- Reticular cells—fibroblasts

Mucoid tissue has an abundance of ground substance composed chiefly of ……

- Hyaluronic acid

The cells in Mucoid Connective Tissue are mainly ……..

- fibroblasts

Mucoid Connective Tissue present in ……

- umbilical cord

…… and ……. are supportive C.T

- bone—cartilage

Enumerate

components of the amorphous material or Ground Substance or Amorphous Matrix

- macromolecules (Glycosaminoglycan—Proteoglycans—Adhesive glycoproteins)
- Soluble proteins of the matrix
- Water

three families of macromolecules of the amorphous material

- Glycosaminoglycan—Proteoglycans—Adhesive glycoproteins

Types of adipose tissue

- Unilocular (white) adipose tissue
- Multilocular (brown) adipose tissue

Two Functions of adipose tissue

- Act as a **source of energy**
- **Give the body surface its shape**
- Act as **thermal insulation** of the body
- **fills up spaces between other tissues**

N.B in Dense connective tissue , there are fewer cells and a clear predominance of Collagen and elastic fibers.

cartilage

Complete

The cartilage attaches the ribs to the sternum is…….{اسئلة كتاب القسم}

- Costal cartilage(Hyaline)

The cells that produce cartilage matrix are ……. and…….{اسئلة كتاب القسم}

- Chondroblasts—Chondrocytes

Cartilage contains predominantly……. fibers. **{اسئلة كتاب القسم}**

- collagen

fibrocartilage contains……..and…….{اسئلة كتاب القسم}

- collagen fibers type 1 and chondrocytes

chondroblasts are found in the…….{اسئلة كتاب القسم}

- Inner cellular (chondrogenic) layer of Perichondrium

fibrocartilage is found in……..and…….{اسئلة كتاب القسم}

- Symphysis pubis and Intervertebral discs

Hyaline cartilage is found in……..and…….{اسئلة كتاب القسم}

- Epiphyseal plate and Costal cartilage

In hyaline cartilage, the chondrocytes are…….. and are found inside……

{اسئلة كتاب القسم}

- mature cartilage cells –lacuna

Examples of elastic cartilage are…….…….and…….{اسئلة كتاب القسم}

- Ear pinna—Epiglottis

Capsule surrounding chondrocyte is called ……..{اسئلة كتاب القسم}

- territorial matrix

Growth of cartilage is either…… or…….{اسئلة كتاب القسم}

- **Interstitial** growth or **Appositional** growth

……is a connective tissue in which the amorphous intercellular substance is hardened to provide rigidity, support and attachment for tissue

- Cartilage

Types of cartilage are …… , ……. and ……

- Hyaline, Yellow elastic and White fibrocartilage

……is a hyaline cartilage which responsible for the longitudinal growth of bone

- Epiphyseal plate

……layer of Perichondrium is Formed of collagen fibers, fibroblasts and blood vessels

- **Outer fibrous**

……layer of Perichondrium is Formed of chondroblast

- **Inner cellular** (or chondrogenic)

Inner cellular layer of Perichondrium is Formed of ……

- chondroblast

chondroblast can be differentiating into ……

- chondrocytes

……and …… are cartilages which devoid the Perichondrium {مهم}

- **articular cartilage** and **white fibrocartilage**

…… is the sulphated glycosaminoglycan in the Matrix of hyaline cartilage

- chondrotin sulphate

…… is the non sulphated glycosaminoglycan in the Matrix of hyaline cartilage

- hyaluronic acid

……is formed of protein core and side chain of glycosaminoglycans

- Protoglycan

Collagen type …… in Hyaline cartilage

- II

Collagen type …….. in White Fibrocartilage

- I

Hyaline cartilage by Hx&E is …… due to the presence of ……..

- basophilic—acidic chondrotin sulphate

cartilage matrix surrounding each chondrocyte called …… matrix

- territorial

Types of Cartilage cells are …… and …… {اسئلة كتاب القسم}

- Chondroblasts—Chondrocytes

……cells are the origin of Chondroblasts

- mesenchymal

Chondroblasts develop from…… cells

- mesenchymal

Chondroblasts are located in …… layer of perichondrium

- Inner cellular (chondrogenic)

……are cartilage forming cells

- Chondroblasts

After maturation , Chondroblasts are surrounded with ……and then they are called……

- lacunae—chondrocytes

chondrocytes Reside in a space in the matrix called ……

- lacuna

Each lacuna is surrounded with a condensed matrix called ……..

- capsule

when lacuna contains group of cells , it is called ……

- cell nest

Elastic fibers can be demonstrated well by ……..

- Orcein stain

Types of Growth of cartilage are ……..and ……

- Interstitial growth—Appositional growth

appositional growth of cartilage occur by ..... {Final2024}

- inner chondrogenic layer of the perichondrium

Interstitial growth of cartilage , ……. Chondrocytes divide and Produce matrix

- Chondrocytes

Enumerate

Types of cartilage {اسئلة كتاب القسم}

- Hyaline cartilage.
- Yellow elastic cartilage.
- White fibrocartilage

Three sites of Hyaline cartilage {مهم} {سؤال في المحاضرة}

خلي بالك برضو ممكن يجي كومبليت ويجيب اي حاجة منهم ويقول نوع الكارتيليج اي

- Epiphyseal plate
- Costal cartilage
- Articular surfaces of joints
- Walls of the large respiratory passages

Three Function of hyaline cartilage

- Support tissue and organs.
- Model for bone development

layers of Perichondrium{مهم}{سؤال في المحاضرة}

- Outer fibrous
- Inner cellular

Three sites of Elastic cartilage {4E} {مهم} {اسئلة كتاب القسم}{سؤال في المحاضرة} {Summar2024}

خلي بالك برضو ممكن يجي كومبليت ويجيب اي حاجة منهم ويقول نوع الكارتيليج اي

- Ear pinna
- Epiglottis

- External auditory meatus
- Eustachian tube

Two sites of White Fibrocartilage **{سؤال في المحاضرة}{مهم}**

خلي بالك برضو ممكن يجي كومبليت ويجيب اي حاجة منهم ويقول نوع الكارتيليج اي

- Symphysis pubis
- Intervertebral discs
- Semilunar cartilage of knee joint
- Around the acetabulum at the tip of glenoid cavity

Two sites in the cartilage which devoid from perichondrium **{سؤال في المحاضرة}**

or what type of cartilage lacks a perichondrium **{اسئلة كتاب القسم}**

- articular cartilage
- white fibrocartilage

Types of cartilage Growth{اسئلة كتاب القسم}

- Interstitial growth
- Appositional growth

Bone

Complete

bone is covered by…… and lined from inside by ……..

periosteum—endosteum

organic material of bone matrix is composed of …… and……..

Collagen type I and Ground substance

Ground substance in bone matrix in the form of …… , ……. , ……. and ……..

- proteoglycan, glycosaminoglycans, osteonectin, and osteocalcin

inorganic material of bone matrix is Formed mainly of calcium in form of ……

- hydroxyapette

……cell is the origin of Osteoprogenitor cells

- mesenchymal

The origin of Osteoblasts is……..cells

- Osteoprogenitor

……. and …… are cells showing negative Golgi image

- plasma cells—osteoblast

Matrix vesicles in osteoblast contain…… enzyme

- alkaline phosphatase

Once the osteoblast becomes surrounded with intercellular substance it is called……

- osteocyte

The function of Osteocytes is …… {Final2024}

- maintain the bone matrix

……called bone forming cells but ……called Bone maintaining cells and ……. called bone eating cells

- osteoblasts—osteocytes – osteoclasts

Bone eating cell is…… {Final2024}

- osteoclast

osteocytes occupy spaces called ……

- lacunae

osteocytes extend cytoplasmic processes that traverse the matrix through……

- canaliculi

Osteoclasts originated from fusion of ……

- monocytes

…….occupy pits called Howship's lacuna

- Osteoclasts

…….creates acidic media in the adjacent matrix substance to dissolve calcium

- Osteoclast

removal of old bone tissue and fractured one by ……..

- osteoclast

……lines the outer surface of the bone

- periosteum

……..lines inner cavities within the bone

- endosteum

The outer surface of compact bone is covered with…….. and bone marrow cavity is lined with ……

- periosteum—endosteum

outer fibrous layer of periosteum composed of …… , …… and ……

- collagen type I, fibroblasts and blood vessels

Inner cellular layer of periosteum composed of ……

- osteoprogenitor cells

Types of bones are …… and …….. {Final2024}

- compact – cancellous (or spongy)

Shaft of long bones represent which type of bone?.....

- **Compact** bone

Each Haversian system is composed of ……. surrounded by regularly arranged concentric lamellae of bone

- Haversian canal

Haversian system also called……

- osteon

Between concentric lamellae, there are……. lamellae

- interstitial

Under the periosteum and endosteum , the lamellae of bone are called the outer and inner ……..lamellae

- circumferential

Cancellous bone is composed of …… of bone separated by bone marrow spaces

- irregular trabeculae

The cells in bone trabeculae are ……

- osteocytes

There is no Haversian systems in……..type of bone

- cancellous

Direct mineralization of matrix secreted by osteoblast called ……..

- **Intramembranous** ossification

Deposition of bone matrix on a preexisting cartilage matrix called ……..

- **Endochondral** ossification

Intra membranous ossification leads to formation of …….. {Final2024}

- spongy bone

Intra cartilaginous ossification leads to formation of ……..{Final2024}

- compact bone

……ossification takes place within a piece of hyaline cartilage

- Endochondral

flat bones as skull and clavicle ossifies by which type of Ossification?.....

- Intramembranous Ossification

Site of Intramembranous Ossification? {Summar2024}

- flat bones as skull, clavicle

long bones ossifies by which type of Ossification?.....

- Endochondral ossification

…….growth of the cartilage of epiphyseal plate continues to increase the length of long bone

- interstitial

The ossification continues of whole bone except at …… and ……..

- articular cartilage—epiphyseal plate

N.B primary ossification center arises in the middle of the shaft of the cartilage model (diaphysis)

N.B secondary ossification centers of occur after birth at the epiphyses

Enumerate

Types of bone cells {سؤال في المحاضرة}

- Osteogenic cells; Osteoblasts, Osteocytes and Osteoclasts

Give short account on osteocytes{سؤال في المحاضرة}

Layers of periosteum{سؤال في المحاضرة}

- outer fibrous layer
- Inner cellular layer

Function of osteoblast

- bone forming cells that secrets the organic component of bone matrix and has a role in the mineralization

Function of osteoclast

- Bone resorption
- Bone remodeling and repair

Types of bone {مهم}

- **Compact** bone
- **Cancellous** bone

Two Sites of Compact bone{مهم} {اسئلة كتاب القسم}

خلي بالك برضو ممكن يجي كومبليت ويجيب اي حاجة منهم ويقول نوعها اي

- Shaft of long bones
- The outer part of the epiphysis
- The outer part of the short and flat bones

Two Sites of Cancellous or spongy bone{مهم}

خلي بالك برضو ممكن يجي كومبليت ويجيب اي حاجة منهم ويقول نوعها اي

- In the center of flat bone and irregular bones.
- At the end of long bones

two ways of Bone formation

- **Intramembranous** ossification
- **Endochondral** ossification

zones which found in the epiphyseal plate{سؤال في المحاضرة}

- Zone of resting cartilage
- Zone of proliferation
- Zone of Hypertrophy
- Zone of Calcification
- Zone of Ossification

Discuss the structure of Haversian system (osteon) {سؤال في المحاضرة}

Discuss the structure of compact bone? {سؤال في المحاضرة}

Discuss the structure of cancellous bone? {سؤال في المحاضرة}

Mcq The outer fibrous layer of periosteum is ……. {Final2024}

A-Dense regular C.T B- Dense irregular C.T

C-thin regular C.T D- thin irregular C.T